STUDENT WORKSHOP FIELD GUIDE | 29 SEPTEMBER 2026

# Overleaf & LaTeX

## for Research Papers and Resumes

A practical guide to writing, compiling, checking, and sharing your work.

LEARN BY BUILDING

## From source code to a checked PDF

Use the same loop for an IEEE paper, a technical report, or a one-page resume.

### RESEARCH PAPER

Organize a question, compare evidence, explain a method, report results.

### OVERLEAF WORKFLOW

Create a project, edit .tex files, recompile, read the log, export the PDF.

### LATEX COMMANDS

Structure sections, equations, figures, tables, citations, and links.

### RESUME

Turn verified details into concise bullets and a clean, editable layout.

### THE WORKING LOOP

Write source > Recompile > Read the PDF > Fix and verify

Prepared for an engineering student workshop | Examples use replaceable placeholders

# 01 The workspace and the working loop

LaTeX is the source language; Overleaf is the browser workspace that compiles it into a PDF.

**LaTeX**
A typesetting system. You write structured source in a plain-text `.tex` file; commands describe headings, math, figures, and references.

**Overleaf**
An online editor for LaTeX projects. It stores project files, runs the compiler, shows the PDF, and supports collaboration.

## The loop you will repeat

Project files → Edit main.tex → Recompile → Read PDF + log → Fix / export

| Task | What to do in Overleaf |
|---|---|
| Start | Create a project from the official target template, or start blank for practice. Give it a clear name. |
| Set up | Keep the main .tex file at the project root. Add a .bib file for sources; make a figures/ folder for images. |
| Compile | Select Recompile. Check the PDF preview after changes; compilation can still report errors even when a preview appears. |
| Debug | Open Logs and output files. Start with the first red error, then inspect the cited line and nearby braces or special characters. |
| Share / export | Use Share to choose access. Download the PDF for reading or the source ZIP to keep an editable copy. History can help recover earlier work. |

### A simple project tree

```
main.tex
references.bib
figures/
  system.png
sections/
  introduction.tex
```

The commands in this guide are typed in the .tex editor; they are not terminal commands. Overleaf runs the compiler when you select Recompile. Button names and plan features can change. Choose the main document, compile often, read the log, and save an exported copy. [1-6]

# 02 The basic LaTeX structure

Every document has a preamble, a body, and one ending command.

```
\documentclass{article}       % document type
\usepackage{graphicx}          % images
\usepackage{amsmath}           % equations
\title{My Technical Report}
\author{Your Name}
\date{}

\begin{document}
\maketitle
\begin{abstract}
State the problem, method, and main result briefly.
\end{abstract}
\section{Introduction}
Start writing here.
\subsection{Motivation}
Explain why the problem matters.
\end{document}
```

## Core commands

| Command | Purpose |
|---|---|
| \documentclass{...} | Select a document class or template. |
| \usepackage{...} | Load an optional feature in the preamble. |
| \begin{...} / \end{...} | Open and close a document environment; names must match. |
| \section{...}, \subsection{...} | Create structured headings. |
| \textbf{...}, \textit{...}, \emph{...} | Bold, italic, or emphasize a short phrase. |
| \label{key}, \ref{key} | Create and use an internal cross-reference. |
| \input{sections/file} | Insert source from another .tex file. |

## Text and punctuation

| In source | Prints | Use |
|---|---|---|
| \% _ \& \$ \# | % _ & $ # | Escape reserved characters in normal text. |
| \texttt{main.tex} | monospace text | Show a filename or short command in prose. |
| blank line | new paragraph | Use \texttt{\\} only when a forced line break is needed. |

Put packages and document settings before \begin{document}. Put the paper's content between \begin{document} and \end{document}. A blank line starts a paragraph; \ is not a paragraph command.

V1

# From LaTeX source to PDF page

The left panel is what you type. The right panel shows the kind of page layout those commands create.

Write the .tex source → Recompile in Overleaf → Read the PDF

**LATEX SOURCE**

```
\section{Introduction}
A short paragraph introduces the problem.
\subsection{Motivation}
Explain why the problem matters.
\begin{itemize}
  \item Define the need
  \item State the objective
\end{itemize}
```

**PAGE PREVIEW**

ILLUSTRATIVE PDF OUTPUT

**1 Introduction**

A clear opening explains the problem and why the reader should care.

**1.1 Motivation**

- Define the need
- State the objective

## How to read the preview

LaTeX turns commands into visual structure: section commands become numbered headings, blank lines separate paragraphs, and list environments create aligned bullets. A PDF preview helps you check hierarchy, spacing, and page flow.

> These small page previews are illustrative drawings made for this guide. Overleaf's exact PDF appearance can vary with the document class, fonts, packages, and compiler settings. Recompile your own source to see its actual output.

# 03 Plan the paper before you format it

A paper is an argument supported by evidence, not just a set of headings.

| Section | What belongs here | Question to answer |
|---|---|---|
| Title + abstract | Topic, problem, method, result, contribution | Can a reader understand the paper quickly? |
| Introduction | Context, problem statement, motivation, objective | What problem are you solving, and why? |
| Related work | Prior methods, findings, limitations | What is already known, and what remains open? |
| Methodology | Data, system, tools, design, evaluation | Could another student follow what you did? |
| Results | Measurements, figures, tables, observations | What did the experiment actually show? |
| Discussion | Meaning, comparison, limitations | How should the results be interpreted? |
| Conclusion | Main takeaway and next steps | What can the reader conclude? |
| References | Every cited source, in the required style | Can each source be found and verified? |

## From topic to research gap

**Topic** → **Search** → **Screen** → **Compare** → **Gap + question**

### A useful comparison note

| Paper / year | Method | Finding | Limitation / relevance |
|---|---|---|---|
| Verified source | What was tested? | What did authors report? | What is missing for your question? |

> A research gap is a specific unresolved limitation or unanswered question in the evidence you reviewed. State the gap, why it matters, and how your study will address it. Do not label a topic as a gap just because you have not searched it yet.

## Query builder

Combine: **topic + method + application + synonyms**. Example from your workshop: `smart traffic AND reinforcement learning AND signal control`. Try alternate terms and record which databases and filters you used.

# 04 IEEE paper starter

Start from the target conference's official template. This example shows where common content goes.

> If you open an IEEE template in Overleaf, keep its document class and preamble. Replace the sample content; do not paste in a second \documentclass line. The venue's current author instructions take priority. [7]

### main.tex - compact working skeleton

```
\documentclass[conference]{IEEEtran}
\usepackage{graphicx}
\usepackage{amsmath}
\usepackage{cite}
\usepackage{url}
\title{Your Research Paper Title}
\author{\IEEEauthorblockN{Your Name}
\IEEEauthorblockA{Department, University\\
City, Country\\ email@example.com}}
\begin{document}
\maketitle
\begin{abstract}
Problem. Method. Main result. Contribution.
\end{abstract}
\begin{IEEEkeywords}
keyword one, keyword two, keyword three
\end{IEEEkeywords}
\section{Introduction}
State the problem and objective. Cite verified work \cite{overleafGuide}.
\section{Related Work}
Compare methods, findings, and limitations.
\section{Methodology}
Describe data, design, and evaluation.
\section{Results and Discussion}
Report results and explain their meaning.
\section{Conclusion}
Summarize the contribution and limitations.
\bibliographystyle{IEEEtran}
\bibliography{references}
\end{document}
```

## What to draft in each part

| Abstract | Introduction | Method + results |
|---|---|---|
| Problem, approach, headline result, contribution. Write it after the main sections. | Motivation, problem statement, research gap, objective, and paper roadmap. | Describe steps and evidence precisely. Separate measured results from interpretation. |

Use `\begin{IEEEkeywords}` for the keyword block in the IEEE conference class. Section names and ordering can vary by venue; follow the selected template. [7]

# 05 Figures, tables, and equations

Use captions and labels so every technical element can be discussed in the text.

## Figures: upload the file first

```
% In the preamble:
\usepackage{graphicx}

% In the body:
\begin{figure}[t]
  \centering
  \includegraphics[width=0.85\linewidth]{figures/system.png}
  \caption{System architecture for the proposed study.}
  \label{fig:system}
\end{figure}

Figure~\ref{fig:system} shows the system architecture.
```

Use a short filename without spaces. Place `\label` after `\caption` so the figure number is captured correctly. [8-9]

## Tables: align columns and label the evidence

```
\begin{table}[t]
  \centering
  \caption{Illustrative evaluation results.}
  \label{tab:results}
  \begin{tabular}{lcc}
    \hline
    Method & Accuracy & Notes \\
    \hline
    Baseline & 0.82 & Reference method \\
    Proposed & 0.87 & Example value only \\
    \hline
  \end{tabular}
\end{table}
```

In `tabular`, & separates cells and `\\` ends a row. Replace the example values with your measured results.

## Equations: inline, display, or aligned

```
Inline: \(x_i^2 + y_i^2 = z_i^2\)

\begin{align}
  J &= \sum_{t=0}^{T} \gamma^t r_t \\
  \text{subject to } & 0 \leq \gamma \leq 1
\label{eq:return}
\end{align}

Equation~\eqref{eq:return} defines the objective.
```

Use [t], [b], or [htbp] as placement suggestions. LaTeX may move a figure or table to keep the page readable. Check the final PDF and the target template's rules. [8-12]

# V2 Four components, side by side

Compare the source with an illustrative preview of how each component can appear in a PDF.

### 1. Figure and caption

**LATEX SOURCE**

```
\includegraphics[width=.8\linewidth]{system.png}
\caption{System architecture.}
\label{fig:system}
% Cite in text: Figure~\ref{fig:system}
```

**PAGE PREVIEW**

ILLUSTRATIVE PDF OUTPUT

Fig. 1. Example system architecture.

### 2. Table with aligned columns

**LATEX SOURCE**

```
\begin{tabular}{lc}
Method & Accuracy \\
Baseline & 0.82 \\
Proposed & 0.87 \\
\end{tabular}
```

**PAGE PREVIEW**

ILLUSTRATIVE PDF OUTPUT

TABLE I. EXAMPLE RESULTS

| Method | Accuracy |
|---|---|
| Baseline | 0.82 |
| Proposed | 0.87 |

*Illustrative values only*

### 3. Numbered equation

**LATEX SOURCE**

```
\begin{equation}
J = \sum_{t=0}^{T} \gamma^t r_t
\label{eq:return}
\end{equation}
% Cite it with \eqref{eq:return}
```

**PAGE PREVIEW**

ILLUSTRATIVE PDF OUTPUT

*J = sum gamma^t r_t*

t = 0, ..., T (1)

### 4. Resume heading and evidence

**LATEX SOURCE**

```
\section*{Projects}
\textbf{Project Title} \hfill 2026
\begin{itemize}
  \item Built a tested project with [tools].
\end{itemize}
```

**PAGE PREVIEW**

ILLUSTRATIVE PDF OUTPUT

**FULL NAME**

City | email@example.com | LinkedIn

**EDUCATION**

- Degree / project, evidence, and date

**PROJECTS**

- Degree / project, evidence, and date

The sample table and resume details are placeholders. Replace them with measured research results and facts you can verify.

# 06 Citations and the .bib file

A citation key connects the sentence in your paper to a verified bibliography entry.

**In main.tex**

```
A related workflow is described in \cite{overleafGuide}.

\bibliographystyle{IEEEtran}
\bibliography{references}
```

Create references.bib in the project and use the citation key exactly as written. Keep the bibliography style that came with the venue template. [13]

**In references.bib - real source used for this syntax example**

```
@misc{overleafGuide,
  author       = {{Overleaf}},
  title        = {Learn LaTeX in 30 minutes},
  howpublished = {\url{https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes}},
  note         = {Accessed 2026-09-28}
}
```

## Reference workflow

Find source → Verify metadata → Export .bib → Upload → Cite + check

| Common problem | Check |
|---|---|
| [?] or missing citation | Does the key in \cite{...} exactly match the key in references.bib? |
| Wrong reference format | Is the template using BibTeX, biblatex, or a venue-specific system? Keep one workflow. |
| Generated entry looks plausible | Open the source record. Verify title, authors, year, venue, pages, and DOI. |

> The example citation above documents the guide syntax; it is not evidence for a research claim. In your paper, cite the original paper or dataset that supports each claim. Never keep a reference just because an AI produced it.

# 07 Use AI in the research workflow

Use tools to find leads and organize reading; use original sources to establish what is true.

| Task | Examples from your directory | Verify by |
|---|---|---|
| Discover papers | Google Scholar, Elicit, Consensus | Open the paper page or publisher record; confirm it exists. |
| Explore connections | ResearchRabbit | Follow citation trails to relevant original studies. |
| Read a source set | NotebookLM, SciSpace | Check any summary against the paper's methods and results. |
| Manage references | Zotero | Review exported .bib fields and citation keys. |
| Draft or code LaTeX | ChatGPT, Gemini | Compile and inspect the PDF; verify every claim and citation. |
| Make a diagram | Mermaid with an AI assistant; Napkin AI | Check labels, arrows, and whether the diagram matches the actual system. |

> These are examples from the supplied tool directory, not endorsements. Product names, access, and features can change. Follow your institution's rules for AI use.

## A prompt that supports a literature review

```
I am studying [research question]. Use only the papers I provide.
Create a comparison table with: citation key, research question,
method, dataset or context, main finding, limitation, and relevance.
For every cell, give a page, section, or table location when possible.
Mark missing information as 'not stated'. Do not invent citations,
results, or a research gap. After the table, suggest 2-3 possible gaps
and explain which evidence supports each suggestion.
```

## Before you write

| Check | Ask yourself |
|---|---|
| Source quality | Is this the original article, a review, a preprint, or a secondary summary? |
| Evidence | Did I read the relevant section, not only the abstract or AI summary? |
| Research gap | Can I point to a limitation, unresolved question, or missing comparison in the sources? |
| Citation | Can I find the source by title, author, year, venue, and DOI? |

# 08 Build resume content before LaTeX

For a student industry application, aim for clear, relevant evidence. Follow the conventions of the target market and role.

## A practical section order

| Section | Include |
|---|---|
| Header | Name, professional email, phone if appropriate, city/region, LinkedIn, GitHub, portfolio. |
| Education | Degree, institution, expected graduation, selected coursework only when relevant. |
| Skills | Tools and methods you can explain and use. Group them by type. |
| Projects / experience | Your role, what you built or analyzed, methods/tools, and a measured result if available. |
| Research / certificates / leadership | Add only sections that strengthen this application. |

## A strong bullet pattern

Action verb + what you built or changed + method/tools + evidence of impact. Example pattern: 'Built [system] with [tools]; evaluated it on [test or dataset]; achieved [measured result].' Use a metric only when you can explain how it was measured.

## Prompt for truthful resume drafting

```
Turn only the facts below into concise resume bullets for [target role].
Use action verbs and name the tools I actually used. Keep the meaning
accurate. Do not invent metrics, responsibilities, skills, dates, or
certifications. If a useful result is missing, ask me what evidence I
have instead of filling it in. Return: (1) 3 bullet options, (2) any
questions needed to make them more specific, and (3) a short list of
claims I should verify. Facts: [paste notes]
```

The resume/CV distinction varies across regions and employers. Treat 'one-page industry resume vs. fuller academic CV' as a workshop starting point, then use the language and length requested by the actual application.

# 09 A clean one-page resume starter

This uses basic LaTeX commands so you can see the structure. Replace every bracketed field with true details.

```
\documentclass[10pt,a4paper]{article}
\usepackage[margin=0.65in]{geometry}
\usepackage[hidelinks]{hyperref}
\usepackage{enumitem}
\setlist[itemize]{leftmargin=*,nosep}
\pagestyle{empty}
\begin{document}
\begin{center}
  {\LARGE \textbf{Full Name}}\\
  City, Country | \href{mailto:name@example.com}{name@example.com}\\
  \href{https://linkedin.com/in/username}{LinkedIn} |
  \href{https://github.com/username}{GitHub}
\end{center}
\section*{Education}
\textbf{Degree}, University \hfill Expected YYYY\\
Relevant coursework: [course], [course]
\section*{Technical Skills}
\textbf{Languages:} [verified skills]\\
\textbf{Tools:} [tools used in projects]
\section*{Projects}
\textbf{Project Title} \hfill Month YYYY
\begin{itemize}
  \item Built [what] using [tools]; tested with [method]; result: [evidence].
\end{itemize}
\section*{Experience}
\textbf{Role}, Organization \hfill Month YYYY -- Month YYYY
\begin{itemize}
  \item [Action] + [scope] + [tool] + [verified outcome].
\end{itemize}
\section*{Achievements}
\begin{itemize}
  \item [Award or responsibility, issuer, date].
\end{itemize}
\end{document}
```

| Command | Why it appears |
|---|---|
| \section*{...} | An unnumbered heading for a compact resume. |
| \hfill | Pushes a date or short detail to the right on the same line. |
| \href{URL}{label} | Creates a clickable email, portfolio, or profile link. |
| \begin{itemize} ... \item | Creates concise achievement bullets. |

A template may define its own commands, margins, and section macros. If you start from a resume template, keep its structure and edit the content first. Compile, check page count, and review every link and date.

# 10 Debug LaTeX without guessing

Fix one cause, recompile, and inspect the result after each meaningful change.

Recompile → Open log → Read first error → Fix source → Recheck PDF

| Message / symptom | Likely cause | First check |
|---|---|---|
| Undefined control sequence | Misspelled command or missing package | Check the command spelling and whether the required package is loaded. |
| Missing $ inserted | Math symbol in normal text | Put math in \( ... \) or escape a text underscore as _. |
| Misplaced alignment tab & | An unescaped ampersand in prose | Write \& in text; keep raw & only as a table or alignment separator. |
| File not found | Image path or capitalization mismatch | Check the project file tree and use the exact filename/path. |
| Citation undefined | Key mismatch or bibliography setup issue | Match the key exactly; inspect references.bib and the template's bibliography commands. |
| ?? in a reference | Label mismatch or unfinished cross-reference pass | Check spelling and placement; recompile and inspect the PDF again. |
| Overfull \hbox | A line is too wide | Rewrite or break a long phrase; check URLs and long technical terms. |

## Safe AI debugging prompt

```
Explain the first LaTeX error below in plain language. Inspect the
source around the reported line. Propose the smallest change that fixes
the cause. Do not rewrite unrelated content. Tell me what to recompile
and what to inspect in the PDF. Error log: [paste first error]
Nearby source: [paste 10-20 lines]
```

> Warnings are not all fatal, but inspect any warning that affects layout, citations, labels, or missing files. A PDF preview alone does not prove the build is clean. Use the log and verify the exported PDF. [3, 14]

# 11 Command quick reference

Keep this page beside Overleaf while you work.

| Need to... | Use |
|---|---|
| Choose class / add feature | \documentclass{article} \| \usepackage{graphicx} |
| Add a heading | \section{...} \| \subsection{...} \| \section*{...} |
| Format text | \textbf{...} \| \textit{...} \| \emph{...} |
| Make a list | \begin{itemize} \item ... \end{itemize} |
| Insert a figure | \includegraphics[width=0.8\linewidth]{figures/name.png} |
| Add caption + internal link | \caption{...} \label{fig:name} \| \ref{fig:name} |
| Write math | \(...\) \| \begin{equation}...\end{equation} \| \begin{align}...\end{align} |
| Cite a source | \cite{key} \| \bibliographystyle{IEEEtran} \| \bibliography{references} |
| Add a hyperlink | \href{https://...}{visible label} |
| Split a long project | \input{sections/introduction} |
| Print reserved symbols | \% _ \& \$ \# \{ \} |

## Before you export

| Research paper | Resume |
|---|---|
| [ ] Question and gap are specific | [ ] Target role and relevant sections are clear |
| [ ] Claims match the cited sources | [ ] Every skill, date, and result is true |
| [ ] Figures/tables/equations have labels | [ ] Contact details and links work |
| [ ] References and template are correct | [ ] Layout fits the requested page length |
| [ ] Compile log checked; PDF re-read | [ ] PDF re-read after final compile |

When using a venue template, its class files and author instructions take priority over generic examples in this guide. For a resume, the job or institution's requested format takes priority.

# 12 Official references and next steps

The linked documentation backs the Overleaf workflow and command examples in this guide.

[1] Overleaf - Learn LaTeX in 30 minutes
[2] Overleaf Docs - The Main document
[3] Overleaf Docs - Recompiling your project
[4] Overleaf Docs - Sharing a project
[5] Overleaf Docs - Downloading a project
[6] Overleaf Docs - History and versioning
[7] IEEE Author Center - Authoring tools and templates
[8] Overleaf - Inserting Images
[9] Overleaf - Cross referencing sections, equations and floats
[10] Overleaf - Tables
[11] Overleaf - Aligning equations with amsmath
[12] Overleaf - Positioning images and tables
[13] Overleaf - Bibliography management with BibTeX
[14] Overleaf Docs - Fixing LaTeX errors

## A short practice sequence

| 1. Open | Create an Overleaf project from the official IEEE conference template or a resume template. |
|---|---|
| 2. Build | Add one section, one citation, one figure or table, and one equation. |
| 3. Check | Recompile, resolve errors, inspect the PDF, and verify each source and claim. |
| 4. Export | Download the PDF and keep the source project so you can edit it later. |

Good technical writing is clear about the problem, method, evidence, and limits. Good LaTeX work compiles cleanly. Good AI-assisted work is checked against the sources and your own experience.